ZERO-TO-JOB-READY

DATA ANALYST 2026

M10 - AI-ASSISTED ANALYTICS

Prompt -> draft -> verify -> source/privacy check -> correct -> human-owned decision

PROPOSE -> VERIFY -> JUDGE

AI can accelerate the workflow; it does not own the evidence.

This book is your teacher

- Assume you can use AI sometimes, but never assume a particular paid plan, add-in, connector or workspace is available.
- The transferable skill is not prompting tricks. It is approved input + clear task + independent verification + source/privacy control + human judgment.
- Every lesson has a direct AIA0-AIA9 worksheet. Protected review opens only after a genuine attempt; changed retry proves transfer.
- Product capabilities and settings change. Current official references are provided for awareness, but employer/client governance and independent verification remain the control layer.
- No AI product is required for protected Gates: the Gate supplies an intentionally flawed AI draft that you must audit.

Normal M10 loop

1. Define decision/rule
2. Minimize approved input
3. Ask AI for plan/draft only
4. Inspect source/grain/denominator
5. Verify material claims independently
6. Correct overclaim/privacy/source issue
7. Save durable evidence
8. Human owner decides

Contents / Index

Module prerequisite and tool map

0. Prerequisite

M00-M09 competency: decision framing, metric/grain definitions, Excel/SQL/Power BI/Python validation, statistics/causal caution and business communication. No AI product expertise is assumed.

Files / tools

- C1_M10_AI_ASSISTED_ANALYTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx - direct AIA0-AIA9 tasks, current references, evidence and mastery.
- Any available AI assistant may be used only where the lesson permits. For Gates, no AI account is required.
- Protected practice/Gate review PDFs - open after a saved attempt only.
- Gate A/B/C workbooks contain synthetic Source_Data plus intentionally flawed AI_Draft.
- Current references are awareness aids. Never copy employer/client secrets or personal data into a tool without approved governance.

AIA0 routing defect repaired

The inherited practice workbook had no direct AIA0 learner sheet and used old P-AIA/M-AIA labels.

RC1 provides one direct AIA0-AIA9 worksheet while preserving the useful audit cases.

No lesson depends on access to a specific commercial AI surface.

Exact practice and Gate route

- AIA0 tool/risk boundary -> AIA1 prompt design -> AIA2 output/file audit -> AIA3 code audit -> AIA4 plans/hypotheses -> AIA5 source hierarchy -> AIA6 privacy -> AIA7 communication -> AIA8 modern surfaces -> AIA9 integrated workflow.
- Every lesson: learner book -> current reference/visual if useful -> guided worksheet -> changed retry -> independent verification -> own-words judgment -> append evidence.
- AIA9 is the integrated module project. Its final analysis must remain reproducible if the AI transcript disappears.
- Gate A is normal; a clean pass ends M10.
- Gate B/C are fresh remediation full retests only after prior failure + saved attempt + protected review + targeted repair + explicit assignment.

Critical failures

Fabricated verification; unsafe data sharing; material numeric/grain/denominator error;
validation control removed to make code run; irrelevant/invented source accepted; unsupported
causal claim; or 'AI said so' used as evidence can block a pass.

AIA0 - What AI is for in analyst work

Open exactly: AIA0_Tool_Risk_Matrix in C1_M10_AI_ASSISTED_ANALYTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

AI can make an analyst faster and also make a wrong answer look polished. This lesson teaches one control that keeps speed separate from evidence ownership.

2. What you are learning

Decide where AI can accelerate analyst work, where independent verification is mandatory, and where privacy/risk/human approval should stop delegation.

3. What you already need / prerequisite

No AI-product skill is assumed.

4. Picture it in your head

Treat AI like a very fast junior assistant who can propose drafts but has no authority to define the business, approve data sharing, certify calculations or make the final decision.

5. New words before we use them

Probabilistic = output is generated from patterns, not guaranteed truth. Draft = starting proposal. Verification = independent check. Human approval = accountable person authorizes action. High-impact decision = outcome can materially affect people/money/access.

6. Watch / visual source

BOOK-LED. No external media is required. The learner book + AIA0_Tool_Risk_Matrix are the complete teacher/practice route.

AIA0 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Classify five tasks. 'Explain a SQL error' -> useful draft + test query. 'Define Active Customer' -> AI may suggest wording, but owner/document controls meaning. 'Paste API key for debugging' -> stop; do not share secret. 'Rewrite verified memo' -> allowed with numbers/limitations locked. 'Choose who loses a benefit' -> analyst/organization policy and human review own the decision.

8. What you should see / be able to say

You can say: AI may assist with mechanics, but the analyst remains accountable for approved input, definition, validation and final judgment.

9. Why it worked

The AI role is bounded before use, and the final claim is tested against a source/rule/check that does not depend on the same unverified response. The durable evidence remains understandable outside the chat.

10. Common beginner mistakes

Equating speed with correctness; treating AI confidence as evidence; letting AI invent a business rule; delegating sensitive decisions; sharing data first and thinking about privacy later.

11. If you get stuck - recovery ladder

Write four columns: task -> what AI may propose -> what must be independently verified -> what requires human/policy approval. If any column is unclear, do not delegate yet.

AIA0 - Do alone, validate and master

12. Now you do a small version alone

Changed case: a manager wants an AI summary of customer complaints plus a recommendation for refunds. Decide what can be summarized, what data should be minimized/redacted, what needs verification and who approves refunds.

13. Validate your result

Your classification must name at least one allowed assist, one independent check, one privacy/permission boundary and one human-owned decision.

14. Teach it back in your own words

Explain PROPOSE -> VERIFY -> JUDGE in your own words and give one example where AI should not own the final decision.

15. Protected review + changed fresh retry

Save the AIA0 worksheet/evidence first. If repair is needed, open the protected AIA0 review, diagnose the weak control, close review, then complete the changed task without using answer material. Keep both attempts.

16. Evidence / mastery + exact next route

Save the completed AIA0_Tool_Risk_Matrix row set + one changed scenario. PASS -> AIA1.

AIA1 - Prompt design: outcome, context, constraints and verification

Open exactly: AIA1_Prompt_Lab in C1_M10_AI_ASSISTED_ANALYTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

AI can make an analyst faster and also make a wrong answer look polished. This lesson teaches one control that keeps speed separate from evidence ownership.

2. What you are learning

Write compact prompts that state the decision/outcome, approved context, definitions/grain, constraints, output shape and the verification you will perform.

3. What you already need / prerequisite

Prerequisite: AIA0 evidence plus the earlier analyst validation habits.

4. Picture it in your head

A prompt is a work order. A vague work order creates rework; a useful one names the job, approved materials, rules, deliverable and inspection checklist.

5. New words before we use them

Outcome = decision/result needed. Context = relevant background. Constraint = rule the output must obey. Grain = what one row represents. Verification plan = checks performed outside the AI answer. Scope = included files/time/entities.

6. Watch / visual source

CURRENT_OFFICIAL_REFERENCE. OpenAI Academy - Analyzing data with ChatGPT | Data analysis directly in ChatGPT + Tips for success. Why: Use current approach-first/file-analysis guidance while the course adds an explicit verification contract. Internal fallback: AIA1 learner book + AIA1_Prompt_Lab. Additional current references: OpenAI Help - Data analysis with ChatGPT.

Open 1 Open 2

AIA1 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Weak: 'Analyze this spreadsheet.' Stronger: 'Using Orders and Targets only, identify which region needs intervention next month. Orders are one row/order; revenue excludes Cancelled; targets are region-month. Return a 5-row table + three observations. Do not infer causes. Show row counts and metric formulas; I will independently recalc top-region revenue, return rate and target variance.'

8. What you should see / be able to say

You should be able to point to the decision, source scope, definitions, prohibited overclaim, output format and independent checks without making the prompt unnecessarily long.

9. Why it worked

The AI role is bounded before use, and the final claim is tested against a source/rule/check that does not depend on the same unverified response. The durable evidence remains understandable outside the chat.

10. Common beginner mistakes

Prompt contains every detail except the decision; no approved-source boundary; missing denominator; asking AI to 'verify itself'; demanding certainty; accepting a fancy output format as evidence.

11. If you get stuck - recovery ladder

Underline six elements: outcome, source/scope, grain/definition, constraints, output, independent checks. Add only the missing element.

AIA1 - Do alone, validate and master

12. Now you do a small version alone

Write a prompt for 'Which support queue needs intervention next week?' Include queue grain, backlog definition, time window, no-causal-overclaim rule, output format and three checks.

13. Validate your result

A reviewer can reproduce the intended metric/scope without asking what 'active', 'completed' or the denominator means.

14. Teach it back in your own words

Explain why a shorter prompt with a precise decision and verification plan can be stronger than a long prompt full of irrelevant context.

15. Protected review + changed fresh retry

Save the AIA1 worksheet/evidence first. If repair is needed, open the protected AIA1 review, diagnose the weak control, close review, then complete the changed task without using answer material. Keep both attempts.

16. Evidence / mastery + exact next route

Save prompt v1, revised prompt, three verification checks and changed prompt. PASS -> AIA2.

AIA2 - Verify AI analysis and file-based answers

Open exactly: AIA2_Output_Audit in C1_M10_AI_ASSISTED_ANALYTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

AI can make an analyst faster and also make a wrong answer look polished. This lesson teaches one control that keeps speed separate from evidence ownership.

2. What you are learning

Audit AI answers about files/data by independently checking material numbers, filters, denominators, row counts and transformations.

3. What you already need / prerequisite

Prerequisite: AIA1 evidence plus the earlier analyst validation habits.

4. Picture it in your head

An AI answer is a claim sheet. You do not ask the same witness whether it is truthful; you compare the claim to source rows, formulas/code or another independent route.

5. New words before we use them

Material claim = number/statement able to change decision. Independent check = verification not dependent on the same unverified output. Reconciliation = agreement between separate calculation paths. Denominator = eligible base of a rate.

6. Watch / visual source

CURRENT_OFFICIAL_REFERENCE. OpenAI Help - File Uploads FAQ | File/spreadsheet analysis, limits and retention awareness. Why: Understand current file-analysis surface and why product capability is not proof of analytical correctness. Internal fallback: AIA2 learner book + AIA2_Output_Audit.

Open 1

AIA2 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Audit: AI says East revenue 1,890,000; controlled source says 1,802,000 -> FAIL numeric. AI says return rate 12.0%; approved definition returned/completed gives 9.6% -> FAIL definition/denominator. AI says West 420 orders; source count 398 -> FAIL count/filter. Record claim, check, evidence and correction separately.

8. What you should see / be able to say

You should never write only 'AI was wrong'. You can state exactly which claim failed, what independent check was used, the corrected value/logic and whether the error is decision-material.

9. Why it worked

The AI role is bounded before use, and the final claim is tested against a source/rule/check that does not depend on the same unverified response. The durable evidence remains understandable outside the chat.

10. Common beginner mistakes

Asking AI to re-answer as the only check; verifying only one total while filters remain wrong; using the AI's own cited calculation as independent evidence; changing denominator after seeing result.

11. If you get stuck - recovery ladder

Freeze the claim. Write source, scope, formula/denominator, expected row count, second calculation path. Recalculate smallest material component first.

AIA2 - Do alone, validate and master

12. Now you do a small version alone
Fresh draft claims North revenue 2,015,000, cancellation rate 7.0%, 515 active accounts. Design checks before deciding whether any claim is usable.

13. Validate your result
Each material claim has a separate source/formula/filter check and a correction status. At least one check must be reproducible without the AI transcript.

14. Teach it back in your own words
Explain why 'the answer looks reasonable' and 'the AI showed code' are not independent validation.

15. Protected review + changed fresh retry
Save the AIA2 worksheet/evidence first. If repair is needed, open the protected AIA2 review, diagnose the weak control, close review, then complete the changed task without using answer material. Keep both attempts.

16. Evidence / mastery + exact next route
Save AIA2_Output_Audit + changed audit. PASS -> AIA3.

AIA3 - AI-generated Excel, SQL, Python and DAX

Open exactly: AIA3_Code_Audit in C1_M10_AI_ASSISTED_ANALYTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

AI can make an analyst faster and also make a wrong answer look polished. This lesson teaches one control that keeps speed separate from evidence ownership.

2. What you are learning

Verify AI-generated Excel, SQL, Python and DAX by defining the business rule first and testing boundaries, grain, joins, missing values and denominators.

3. What you already need / prerequisite

Prerequisite: AIA2 evidence plus the earlier analyst validation habits.

4. Picture it in your head

Syntax is the spelling of the instruction; business logic is what the instruction means. AI can spell perfectly and still implement the wrong rule.

5. New words before we use them

Boundary test = just below/exactly/above threshold. Join cardinality = allowed matches between tables. Result grain = what one output row represents. Validation control = assertion/check designed to fail when logic is unsafe.

6. Watch / visual source

OPTIONAL_VISUAL_COMPANION. Alex The Analyst - Learn SQL Beginner to Advanced in Under 4 Hours | Use only for SQL syntax/debugging visuals. Why: Useful syntax refresher; not the primary AI-code-verification lesson. Internal fallback: AIA3 learner book + multi-tool code audit.

Open 1

AIA3 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Excel: Eligible if Score >= 90; test 89/90/91. SQL: DISTINCT after joining order_lines does not prove one row/order—aggregate or validate grain. Python: dropna() may hide unknown product IDs—keep/review unmatched. DAX: Returns/All Orders violates returned/completed denominator—fix definition first.

8. What you should see / be able to say

You can reject code that runs if it violates a business rule, and you can design a tiny test that exposes the error.

9. Why it worked

The AI role is bounded before use, and the final claim is tested against a source/rule/check that does not depend on the same unverified response. The durable evidence remains understandable outside the chat.

10. Common beginner mistakes

Removing validate/assertion because it raises; trusting SELECT DISTINCT as grain proof; blanket dropna/fillna; testing only happy paths; accepting formula syntax without denominator review.

11. If you get stuck - recovery ladder

Stop reading code. Write the business rule in one sentence, create 2-3 boundary/grain examples, then run the smallest test against the draft.

AIA3 - Do alone, validate and master

12. Now you do a small version alone

Create one changed mini-case per tool where plausible-looking AI code violates a rule. Design one test per case that would catch it.

13. Validate your result

All four cases name the rule, expected behavior and objective test. No case is considered fixed merely because it executes.

14. Teach it back in your own words

Explain 'valid syntax is not valid analysis' with one Excel/SQL/Python/DAX example.

15. Protected review + changed fresh retry

Save the AIA3 worksheet/evidence first. If repair is needed, open the protected AIA3 review, diagnose the weak control, close review, then complete the changed task without using answer material. Keep both attempts.

16. Evidence / mastery + exact next route

Save AIA3_Code_Audit + changed cases. PASS -> AIA4.

AIA4 - AI-assisted analysis: plan, hypotheses and exploration

Open exactly: AIA4_Analysis_Plan in C1_M10_AI_ASSISTED_ANALYTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

AI can make an analyst faster and also make a wrong answer look polished. This lesson teaches one control that keeps speed separate from evidence ownership.

2. What you are learning

Use AI to broaden an analysis plan and hypotheses without turning suggestions into facts or endlessly slicing data.

3. What you already need / prerequisite

Prerequisite: AIA3 evidence plus the earlier analyst validation habits.

4. Picture it in your head

AI is a brainstorming wall: it can put many possible explanations on sticky notes. The analyst labels which are facts, assumptions or hypotheses and chooses which few are worth testing.

5. New words before we use them

Fact = supported by evidence. Assumption = provisional rule/belief. Hypothesis = testable possible explanation. Testability = evidence can discriminate. Impact = would change decision. Stop rule = condition to stop further slicing.

6. Watch / visual source

CURRENT_OFFICIAL_REFERENCE. OpenAI Academy - Analyzing data with ChatGPT | Ask for an approach; anomalies and hypotheses. Why: Use AI to widen possible analyses while keeping facts, assumptions and hypotheses separate. Internal fallback: AIA4 learner book + AIA4_Analysis_Plan.

Open 1

AIA4 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Renewal fell 72% to 65%. Generate six possible drivers using plan/region/tenure/support/payment/onboarding. Mark each as hypothesis unless already evidenced. Rank by impact x testability. Start with payment failures/onboarding if measured; defer vague 'market sentiment'. Stop when top actionable segments/drivers are stable enough for the decision.

8. What you should see / be able to say

Your plan is a short ranked queue with required evidence and stop rule—not a list of every column combination.

9. Why it worked

The AI role is bounded before use, and the final claim is tested against a source/rule/check that does not depend on the same unverified response. The durable evidence remains understandable outside the chat.

10. Common beginner mistakes

AI suggestion written as finding; six hypotheses treated equally; endless segmentation until something dramatic appears; causal wording before testing; no decision link.

11. If you get stuck - recovery ladder

For each proposed item ask: What evidence would prove/disprove it? Would the answer change the decision? If no, deprioritize.

AIA4 - Do alone, validate and master

12. Now you do a small version alone

Complaint rate doubled. Produce six hypotheses, reject/defer two weak ones and define a decision-based stop rule.

13. Validate your result

Every proposed explanation is labeled fact/assumption/hypothesis and has an evidence/testability note. Top priorities are justified.

14. Teach it back in your own words

Explain why AI is useful for breadth but not authority in root-cause analysis.

15. Protected review + changed fresh retry

Save the AIA4 worksheet/evidence first. If repair is needed, open the protected AIA4 review, diagnose the weak control, close review, then complete the changed task without using answer material. Keep both attempts.

16. Evidence / mastery + exact next route

Save ranked AIA4 plan + changed symptom plan. PASS -> AIA5.

AIA5 - Hallucinations, citations and source hierarchy

Open exactly: AIA5_Source_Hierarchy in C1_M10_AI_ASSISTED_ANALYTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

AI can make an analyst faster and also make a wrong answer look polished. This lesson teaches one control that keeps speed separate from evidence ownership.

2. What you are learning

Match each claim to the smallest authoritative evidence capable of verifying it, and reject irrelevant or fabricated citations.

3. What you already need / prerequisite

Prerequisite: AIA4 evidence plus the earlier analyst validation habits.

4. Picture it in your head

Evidence is a key: the right key must fit the exact lock. A real citation about the topic can still be the wrong key for your specific claim.

5. New words before we use them

Source hierarchy = preferred evidence order by claim type. Authoritative = source with appropriate ownership/direct evidence. Primary source = original data/document/provider. Freshness = how current the evidence must be. Citation relevance = supports exact claim, not merely topic.

6. Watch / visual source

BOOK-LED. No external media is required. The learner book + AIA5_Source_Hierarchy are the complete teacher/practice route.

AIA5 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

File contains 18,240 completed orders -> source file/query. Company definition changed in July -> approved business owner/data dictionary/change log. Current product supports feature -> current official provider docs. 'Onboarding caused renewal rise' -> experiment/quasi-experimental evidence; a product page or blog cannot prove causality.

8. What you should see / be able to say

You can explain not only which source you chose but why another plausible citation is insufficient for that claim.

9. Why it worked

The AI role is bounded before use, and the final claim is tested against a source/rule/check that does not depend on the same unverified response. The durable evidence remains understandable outside the chat.

10. Common beginner mistakes

Citation count as quality; old docs for changing product facts; blog used for internal business definition; source confirms a related topic but not exact number; citation treated as causal proof.

11. If you get stuck - recovery ladder

Classify the claim: file/internal definition/current product/causal/other. Then choose the closest primary/authoritative evidence and check date/scope.

AIA5 - Do alone, validate and master

12. Now you do a small version alone

For contract rule, current software capability, internal revenue total and causal marketing claim, choose best evidence and one insufficient citation type.

13. Validate your result

Every claim has evidence with matching authority, scope and freshness. Unsupported claims remain labelled unsupported.

14. Teach it back in your own words

Explain why a citation can be real and still fail verification.

15. Protected review + changed fresh retry

Save the AIA5 worksheet/evidence first. If repair is needed, open the protected AIA5 review, diagnose the weak control, close review, then complete the changed task without using answer material. Keep both attempts.

16. Evidence / mastery + exact next route

Save AIA5_Source_Hierarchy + changed claim set. PASS -> AIA6.

AIA6 - Privacy, confidentiality and data minimization

Open exactly: AIA6_Privacy_Minimization in C1_M10_AI_ASSISTED_ANALYTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

AI can make an analyst faster and also make a wrong answer look polished. This lesson teaches one control that keeps speed separate from evidence ownership.

2. What you are learning

Make privacy/confidentiality/data-minimization decisions before using AI, with employer/client governance taking precedence over convenience.

3. What you already need / prerequisite

Prerequisite: AIA5 evidence plus the earlier analyst validation habits.

4. Picture it in your head

Before asking 'what prompt?', ask 'what am I allowed to disclose?' Imagine packing only the minimum approved fields into a sealed envelope; secrets/PII stay out unless explicitly governed and required.

5. New words before we use them

Data minimization = share only what is necessary. Redaction = remove sensitive fields. Synthetic data = artificial example with no real person's record. Credential = secret used to access system. Governance = organization rules/approvals. PII = information identifying a person.

6. Watch / visual source

CURRENT_OFFICIAL_REFERENCE. OpenAI Help - Data Controls FAQ | Current ChatGPT data-control awareness. Why: Know consumer controls without treating them as employer/client approval. Internal fallback: AIA6 learner book + privacy matrix. Additional current references: OpenAI Help - How your data is used to improve model performance.

Open 1 Open 2

AIA6 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Customer names/emails/phones for a formula bug -> do not paste by default; reproduce with synthetic/minimized fields or approved environment. Salary sheet -> ask syntax using column names/sample, not salaries. Public synthetic course data -> acceptable course case. API key -> secret; revoke/rotate if exposed and do not share it.

8. What you should see / be able to say

Your answer names allowed/not-allowed data, minimum substitute, required approval/policy and what to do with credentials/secrets.

9. Why it worked

The AI role is bounded before use, and the final claim is tested against a source/rule/check that does not depend on the same unverified response. The durable evidence remains understandable outside the chat.

10. Common beginner mistakes

Assuming 'training off' means employer approval; uploading whole file for one formula; masking names but leaving emails/IDs; pasting tokens/passwords; treating anonymization as automatic.

11. If you get stuck - recovery ladder

Stop upload. List fields needed for the task, fields removable, sensitive/secret fields, approved environment/policy, and synthetic alternative.

AIA6 - Do alone, validate and master

12. Now you do a small version alone

Decide for a non-public contract, anonymized aggregate, complaint text and database password. State minimum data and approval route.

13. Validate your result

No credential/secret is shared; sensitive/confidential data has a justified approved path or is minimized/synthesized. Product settings are not used as the only governance evidence.

14. Teach it back in your own words

Explain the difference between a product data-control setting and permission to share employer/client data.

15. Protected review + changed fresh retry

Save the AIA6 worksheet/evidence first. If repair is needed, open the protected AIA6 review, diagnose the weak control, close review, then complete the changed task without using answer material. Keep both attempts.

16. Evidence / mastery + exact next route

Save AIA6_Privacy_Minimization + changed scenarios. PASS -> AIA7.

AIA7 - AI-assisted communication and documentation

Open exactly: AIA7_Communication_Log in C1_M10_AI_ASSISTED_ANALYTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

AI can make an analyst faster and also make a wrong answer look polished. This lesson teaches one control that keeps speed separate from evidence ownership.

2. What you are learning

Use AI to rewrite/structure communication while preserving verified numbers, uncertainty, causal boundaries and a material change log.

3. What you already need / prerequisite

Prerequisite: AIA6 evidence plus the earlier analyst validation habits.

4. Picture it in your head

Think of AI as an editor holding a red pen, not a journalist allowed to invent new facts. The verified evidence is locked; wording may improve but facts cannot expand.

5. New words before we use them

Change log = record of substantive edits. Evidence lock = numbers/scope/limitations that must not change. Audience = decision-maker/readership. Overclaim = wording stronger than evidence. Human-owned conclusion = final message approved by analyst.

6. Watch / visual source

BOOK-LED. No external media is required. The learner book + AIA7_Communication_Log are the complete teacher/practice route.

AIA7 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Verified: late-delivery 14.2%, East 21.8%, Jan-Jun only. AI rewrite invents poor warehouse staffing and immediate hiring. Correct: preserve exact rates/time scope, say East is higher, state cause is not established, recommend investigation/test rather than guaranteed staffing fix. Log removed causal claim and preserved limitation.

8. What you should see / be able to say

Final communication contains no new number/cause/source and the change log shows what AI changed and what you corrected.

9. Why it worked

The AI role is bounded before use, and the final claim is tested against a source/rule/check that does not depend on the same unverified response. The durable evidence remains understandable outside the chat.

10. Common beginner mistakes

AI rounds material number differently; adds 'significant' without test; drops limitation; converts association into cause; recommendation becomes stronger than evidence.

11. If you get stuck - recovery ladder

Make a locked-facts block first. Compare draft sentence-by-sentence: number, scope, causal strength, recommendation strength, limitation.

AIA7 - Do alone, validate and master

12. Now you do a small version alone
Rewrite the same verified evidence for Finance and reject one unsupported embellishment. Record changes.

13. Validate your result
Every final number matches verified source; limitation remains visible; no causal strength increases; analyst is named final owner.

14. Teach it back in your own words
Explain which parts of an executive memo AI may improve and which parts it may not change without new verification.

15. Protected review + changed fresh retry
Save the AIA7 worksheet/evidence first. If repair is needed, open the protected AIA7 review, diagnose the weak control, close review, then complete the changed task without using answer material. Keep both attempts.

16. Evidence / mastery + exact next route
Save AIA7_Communication_Log + changed audience version. PASS -> AIA8.

AIA8 - Modern analyst AI surfaces: know them, do not depend on them

Open exactly: AIA8_Modern_Surfaces in C1_M10_AI_ASSISTED_ANALYTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

AI can make an analyst faster and also make a wrong answer look polished. This lesson teaches one control that keeps speed separate from evidence ownership.

2. What you are learning

Recognize modern AI surfaces in ChatGPT, spreadsheets and BI while applying the same transferable source, permission, review, validation, approval and rollback controls.

3. What you already need / prerequisite

Prerequisite: AIA7 evidence plus the earlier analyst validation habits.

4. Picture it in your head

AI surfaces are different dashboards on the same safety vehicle. Buttons move, but brakes, mirrors and seatbelts—scope, permissions, validation, approval—remain.

5. New words before we use them

Surface = place AI appears (chat/add-in/BI/agent). Permission = allowed data/action scope. Side effect = system/workbook change. Rollback = undo/restore. Grounding = context/source supplied to model. Availability = plan/platform/tenant dependent.

6. Watch / visual source

ASSIGNED_VISUAL. OpenAI - Introducing ChatGPT for Excel and Google Sheets | Full focused visual. Why: See spreadsheet-native AI in motion; learn the idea, not a button location. Internal fallback: AIA8 learner book + AIA8_Modern_Surfaces. Additional current references: OpenAI - Introducing ChatGPT for Excel and new financial data integrations, OpenAI Help - Creating and editing documents, spreadsheets, and presentations with ChatGPT Work, Microsoft Support - Get started with Copilot in Excel, Microsoft Learn - Use Copilot with Power BI reports and semantic models.

AIA8 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Compare: ChatGPT file analysis -> verify uploaded scope/calculation. Spreadsheet-native AI -> review changed cells/formulas and preserve workbook. Copilot Excel -> inspect edits/plan and permissions. Power BI Copilot -> model/DAX/report validation remains. Agent/connected app -> least-necessary permission + explicit approval for actions.

8. What you should see / be able to say

You can evaluate a new AI product using controls without needing to memorize its current interface or own its paid plan.

9. Why it worked

The AI role is bounded before use, and the final claim is tested against a source/rule/check that does not depend on the same unverified response. The durable evidence remains understandable outside the chat.

10. Common beginner mistakes

Learning buttons instead of controls; assuming access/feature from a video; letting AI edit shared workbook without review; treating semantic-model weakness as AI problem; broad connector permissions.

11. If you get stuck - recovery ladder

Use seven questions: What data? What action? What source? What can change? How verify? Who approves? How rollback? If any answer is unknown, use awareness-only mode.

AIA8 - Do alone, validate and master

12. Now you do a small version alone

Pick any future AI surface and fill the same control matrix without relying on marketing copy as proof of permissions/safety.

13. Validate your result

Your matrix separates capability awareness from actual access and requires verification/approval/rollback for material side effects.

14. Teach it back in your own words

Explain why this lesson should remain useful even if every 2026 button moves.

15. Protected review + changed fresh retry

Save the AIA8 worksheet/evidence first. If repair is needed, open the protected AIA8 review, diagnose the weak control, close review, then complete the changed task without using answer material. Keep both attempts.

16. Evidence / mastery + exact next route

Save AIA8_Modern_Surfaces + changed surface checklist. PASS -> AIA9.

AIA9 - Integrated AI-assisted analyst workflow

Open exactly: AIA9_Integrated_Workflow in C1_M10_AI_ASSISTED_ANALYTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

AI can make an analyst faster and also make a wrong answer look polished. This lesson teaches one control that keeps speed separate from evidence ownership.

2. What you are learning

Run a complete AI-assisted analyst workflow whose final analysis remains reproducible and defensible even if the AI transcript disappears.

3. What you already need / prerequisite

Prerequisite: AIA8 evidence plus the earlier analyst validation habits.

4. Picture it in your head

AI is one temporary station inside a permanent evidence pipeline. The durable deliverable is source + definitions + code/calculations + validation + decision log, not the chat transcript.

5. New words before we use them

Reproducible = another analyst can rerun from approved inputs. Verification log = claim/check/evidence/correction record. Provenance = where result came from. Approval checkpoint = human review before material side effect. Final owner = accountable analyst/stakeholder.

6. Watch / visual source

BOOK-LED. No external media is required. The learner book + AIA9_Integrated_Workflow are the complete teacher/practice route.

AIA9 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Support backlog: frame 'is backlog improving enough to reduce escalation risk?'; define backlog/period; share only synthetic/approved aggregate; ask AI for analysis plan; use it to draft code/formulas/memo; independently verify trend/count/denominator; correct any invented cause; save source path, calculation, checks, corrected conclusion and limitation. Delete transcript mentally: deliverable should still stand.

8. What you should see / be able to say

Every material conclusion can be reconstructed from durable source/calculation/evidence without relying on hidden AI reasoning or transcript.

9. Why it worked

The AI role is bounded before use, and the final claim is tested against a source/rule/check that does not depend on the same unverified response. The durable evidence remains understandable outside the chat.

10. Common beginner mistakes

Chat transcript is only documentation; prompt contains sensitive source data unnecessarily; AI code runs but no independent total; correction made but old output remains in memo; no human approval point.

11. If you get stuck - recovery ladder

Build the evidence chain without AI first: decision -> definitions -> approved source -> required checks -> final owner. Then place AI only in draft/brainstorm/mechanics stages.

AIA9 - Do alone, validate and master

12. Now you do a small version alone

Fresh case: did discounting improve revenue quality? Build the full workflow with minimized context, plan, draft mechanics, independent verification, correction log and human-owned conclusion.

13. Validate your result

Reviewer can reproduce at least two material checks, trace every final claim to durable evidence, see privacy/source decisions and identify final approval owner.

14. Teach it back in your own words

Explain how AI can disappear from the project tomorrow without making your analysis unverifiable.

15. Protected review + changed fresh retry

Save the AIA9 worksheet/evidence first. If repair is needed, open the protected AIA9 review, diagnose the weak control, close review, then complete the changed task without using answer material. Keep both attempts.

16. Evidence / mastery + exact next route

Save AIA9_Integrated_Workflow + verification log. PASS -> Gate A. PARTIAL/FAIL -> targeted repair before Gate A.

End-of-module mastery checklist

- AIA0: Separate AI assistance, independent verification, privacy/permission boundary and human decision ownership.
- AIA1: Prompt states outcome, approved scope, grain/definitions, constraints, output and independent checks.
- AIA2: Recalculate material file/data claims through independent source/formula/filter checks.
- AIA3: Validate AI-generated Excel/SQL/Python/DAX against business rules, boundaries, grain, joins and denominators.
- AIA4: Separate facts/assumptions/hypotheses; rank by impact/testability; use stop rule.
- AIA5: Match claim type to authoritative/current evidence; reject irrelevant citation.
- AIA6: Minimize/redact/synthesize before sharing; product setting does not override employer/client governance.
- AIA7: Preserve verified numbers/scope/limitations and log substantive AI communication changes.
- AIA8: Apply source/permission/review/validation/approval/rollback controls across changing AI surfaces.
- AIA9: Deliver a reproducible workflow whose evidence survives without the AI transcript.

Gate readiness

Attempt Gate A only after AIA9 + verification log are saved and privacy/source/causal controls are explicit.

A clean Gate A pass ends M10; B/C are remediation full retests only.

Revision / AI verification quick reference

PROPOSE	AI may brainstorm, explain or draft.
VERIFY	Check material claim independently against source/rule/test.
JUDGE	Human analyst owns interpretation and decision.
Prompt	Outcome + approved context + definitions/grain + constraints + output + checks.
File answer	Recheck row counts, filters, denominators, transformations and material totals.
Generated code	Business rule first; test boundary/grain/join/missing/denominator.
Source	Evidence authority/freshness must match exact claim type.
Privacy	Minimize/redact/synthesize; never paste credentials/secrets.
Communication	Lock verified facts/limitations; AI cannot add stronger evidence.
Modern surfaces	Ask: data, action, source, change, verify, approve, rollback.
Reproducible	Source + calculations + checks + decision log must survive without transcript.

Next route: AIA9 -> Gate A -> M11 Portfolio/Capstone/Hiring after M10 competency pass.